Chain of Submodules

Definition 0.1. : A **chain of submodules** of M is a sequence

$$M = M_0 \supset M_1 \supset M_2 \supset \cdots \supset M_n = (0)$$

where M_i are submodules of M. (Here, the notation $\supset$ means strictly inclusion)

Composition Series

Definition 0.2. : A **composition series** of M is a chain of submodules of M

$$M = M_0 \supset M_1 \supset \cdots \supset M_n = (0)$$

such that M_i/M_{i+1} is simple, i.e., M_i/M_{i+1} only has trivial submodules. n is called the **length** of the chain.

Example 0.1. : $\mathbb{Z}_4 \supset 2\mathbb{Z}_4 \supset (0)$, $\mathbb{Z}_{12} \supset 3\mathbb{Z}_{12} \supset 6\mathbb{Z}_{12} \supset (0)$.

Theorem 0.1. *: Let M be a module with a composition series of length n. Then every composition series of M is of length n. More over, every chain of submodules of M can be extended to a composition series.*

Proof: Let $l(M)$ denote the minimum of the length of composition series of M, hence $l(M) \leq n$. Let $N \subset M$ be a proper submodule of M, let

$$M = M_0 \supset M_1 \supset M_2 \supset \cdots \supset M_n = (0)$$

be a composition series of M of length n. Then take $N_i := M_i \cap N$, so we can know that

$$N = N_0 \supset N_1 \supset \cdots \supset N_n = (0) \quad (\#)$$

is a chain of submodules of N. Consider the injective homomorphism

$$N_i/N_{i+1} = (M_i \cap N)/(M_{i+1} \cap N) \hookrightarrow M_i/M_{i+1}$$

Since M_i/M_{i+1} is simple and N_i/N_{i+1} is a submodule of M_i/M_{i+1}, hence $N_i/N_{i+1} = (0)$ or $N_i/N_{i+1} = M_i/M_{i+1}$. By deleting the repeating terms in the chain (#), we get a composition series of N. Hence we have $l(N) \leq l(M)$.

If $l(N) = l(M)$, then $N_i/N_{i+1} = (M_i \cap N)/(M_{i+1} \cap N) \neq 0$ for any i, then $N_{n-1}/N_n = M_{n-1}/M_n$ and thus $M_{n-1} = N_{n-1}$ since $M_n = N_n = (0)$. Since $N_{n-2}/N_{n-1} = M_{n-2}/M_{n-1}$, so $N_{n-2} = M_{n-2}$. Repeating the discussion, we can know that $N_i = M_i$ $(\forall i)$. Particularly, we get $N_0 = M_0$, i.e., $N = M$, which is a contradiction. So we know that $l(N) < l(M)$.

Now, let

$$M = M_0 \supset M_1 \supset \cdots \supset M_n = (0)$$

be a composition series of M, then by the discussion above, we get

$$l(M) \geq l(M_1) + 1 \geq l(M_2) + 2 \geq \cdots \geq l(M_n) + n = n$$

thus $l(M) = n$, which means that the length of any compostion series of M is n;

Let

$$M = M_0 \supset M_1 \supset \cdots \supset M_k = (0)$$

be a chain of submodules of M, hence M_{k-1} has a composition series (by the proof above). If M_{k-2}/M_{k-1} is not simple, then M_{k-1} is not a maximal submodule of M_{k-2}. Let M'_{k-2} be a maximal submodule of M_{k-2} containing M_{k-1}, i.e., $M_{k-1} \subset M'_{k-2} \subset M_{k-2}$. Repeat the process like this, we can extend the chain to a composition series. □

Length of Modules

Definition 0.3. :If the module M has a compostion series, then the length of the composition series of M is called **the length of** M, denoted by $l(M)$. (By the propostion above, we know that the length of the module M is unique); If M does not have a composition series, then $l(M) := \infty$.

Proposition 0.1. *: M has a composition series $\Leftrightarrow$ M satisfies a.c.c and d.c.c.*

Proof: ($\Rightarrow$): It's clear that M satisfies a.c.c and d.c.c since every chain of submodules of M can be extended to a composition series;

($\Leftarrow$): Let M_1 be a maximal submodule of M (such M_1 exists since M satisfies a.c.c). Let Σ_1 be the collection of proper submoduls of M_1 and let M_2 be a maximal element in Σ_1, i.e., M_2 is a maximal submodule of M_1. Continuing like this, we get

$$M \supset M_1 \supset M_2 \supset \ldots,$$

Since M satisfies d.c.c, thus $\exists k \in \mathbb{N}$ such that $M_k = (0)$. Hence this chain terminates,

$$M \supset M_1 \supset \cdots \supset M_k = (0)$$

and M_i/M_{i+1} is simple since M_{i+1} is a maximal submodule of M_i, which shows that this chain is a composition series. □

Proposition 0.2. *: Let*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be an exact sequence. If M has a composition sereis, then we have

$$l(M) = l(M') + l(M'')$$

i.e., l is an additive function.

Proof: Consider $M' \hookrightarrow M$ and $M'' \cong M/M'$. Let

$$M'' = M/M'' \supset M_1/M' \supset \cdots \supset M_{r-1}/M' \supset M'/M' = (0)$$

be a composition series of M'', where $M_i \supseteq M'$. Let

$$M' \supset M'_1 \supset \cdots \supset M'_{s-1} \supset M'_s = (0)$$

be a compostion series of M'. Consider

$$M \supset M_1 \supset M_2 \supset \cdots \supset M_{r-1} \supset M_r = M' \supset M'_1 \supset \cdots \supset M'_{s-1} \supset M'_s = (0) \quad (\#)$$

thus $M_i/M_{i+1} \cong (M_i/M'')/(M_{i+1}/M'')$, which is simple. Thus (#) is a composition series of M, hennce $l(M) = l(M') + l(M'')$. □

Example 0.2. : $M = k[x]/(x^2)$, $A = k[x]$. Then $M = k[x]/(x^2) \supset (x)/(x^2) \supset (0)$, hence $l(\frac{k[x]}{(x^2)}) = 2$. Actually, we can know that $l(\frac{k[x]}{(x^n)}) = n$.

Proposition 0.3. *: For a field k, let V be a k-vector space, then TFAE:*
(1) $\dim V < \infty$*;*
(2) V is of finite length (as a k-module);
(3) V satiefies a.c.c;
(4) V satiesfies d.c.c.

Proof (1)⇔(3) and (1)⇔(4) are clear by linear algebra.

(1)⇒(2): Suppose $\{e_1, e_2, \ldots, e_n\}$ is a basis of V, let $V_i := \mathrm{Span}\{e_1, e_2, \ldots, e_{n-i}\}$ $(0 \leq i \leq n)$, then

$$V_0 = V \supset V_1 \supset V_2 \supset \cdots \supset V_n = (0)$$

and V_i/V_{i+1} is a vector space of dimension 1, which implies that V_i/V_{i+1} does not have non-trivial subspace, hence V_i/V_{i+1} is simple. Thus this is a composition series of V and $l(V) < \infty$;

(2)⇒(3) and (2)⇒(4) are clear to see. □

Proposition 0.4. *: Let A be a ring,* $\mathfrak{m}_1, \mathfrak{m}_2, \ldots, \mathfrak{m}_n$ *are maximal ideals of A such that* $\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_n = (0)$*, then A is a Noetherian ring ⇔ A is an Artinian ring.*

Proof: Sine we have

$$A \supseteq \mathfrak{m}_1 \supseteq \cdots \supseteq \mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_{n-1} \supseteq \mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_n = (0)$$

now let

$$\pi_i : \mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_i \to \mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_i/\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_{i+1}$$
$$x \mapsto \overline{x}$$

be the natural homomorphism, hence for $\forall a \in A$, we have $a\overline{x} = \overline{ax}$ $(x \in \mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_i)$. Define

$$A/\mathfrak{m}_{i+1} \times \mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_i/\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_{i+1} \to \mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_i/\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_{i+1}$$

$$(\overline{r}, \overline{x}) \mapsto \overline{rx}$$

then we can check that this scalar multiplication is well-defined and hence $\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_i/\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_{i+1}$ is an $A/\mathfrak{m}_{i+1}$-module. Since $A/\mathfrak{m}_i$ is a field, thus $\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_i/\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_{i+1}$ is a vector space. So by the proposition above, we know that $\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_i/\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_{i+1}$ satiesfies a.c.c $\Leftrightarrow$ $\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_i/\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_{i+1}$ satiesfies d.c.c. Since every $\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_i/\mathfrak{m}_1\mathfrak{m}_2 \ldots \mathfrak{m}_{i+1}$ satisfies a.c.c (resp. d.c.c) $\Leftrightarrow$ A satisfies a.c.c (resp. d.c.c), we can know that A satisfies a.c.c $\Leftrightarrow$ A satisfies d.c.c, i.e., A is a Noetherian ring $\Leftrightarrow$ A is an Artinian ring. □

Proposition 0.5. *: Suppose A is a Noetherian ring, then we have*
(1) Every ideal of A is finitely generated;
(2) For an ideal I of A, then both I and A/I are Noetherian rings;
(3) If A is a subring of B and B is finitely generated as an A-module, then B is a Noetherian ring;
(4) If S is a multiplicative set of A, then $S^{-1}A$ is a Noetherian ring.

Proof: (1): ✓.

(2) Consider the exact sequence

$$0 \longrightarrow I \longrightarrow A \longrightarrow A/I \longrightarrow 0$$

since A is Noetherian, then so are I and A/I.

(3) Since B is a finitely generated A-module, hence we can know that B is Notherian as an A-module. Thus B is also Noetherian as a B-module, which means by definition B is a Noetherian ring.

(4) We have known that the ideal in $S^{-1}A$ is of the form $S^{-1}I$, where I is an ideal in A. Since A is a Noetherian ring, hence every ideal $I \triangleq \langle x_1, x_2, \ldots, x_n \rangle$, thus $S^{-1}I = \langle \frac{x_1}{1}, \frac{x_2}{1}, \ldots, \frac{x_n}{1} \rangle$. Hence $S^{-1}A$ is a Noetherian ring. □